



Tennessee Valley Authority, Sequoyah Nuclear Plant, P.O. Box 2000, Soddy Daisy, Tennessee 37384

September 18, 2018

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

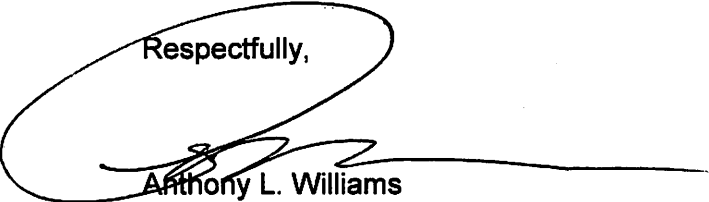
Sequoyah Nuclear Plant, Units 1 and 2
Renewed Facility Operating License Nos. DPR-77 and DPR-79
NRC Docket Nos. 50-327 and 50-328

Subject: Licensee Event Report 50-327 and 50-328/2018-001-00, Inadequate Post Maintenance Testing Results in Condition Prohibited by Technical Specifications

The enclosed licensee event report provides details concerning a situation in which inadequate post maintenance testing resulted in one train of the Emergency Gas Treatment System being inoperable for longer than allowed by Technical Specifications. This report is being submitted in accordance with 10 CFR 50.73(a)(2)(i)(B), as any operation or condition which was prohibited by the plant's Technical Specifications and in accordance with 10 CFR 50.73(a)(2)(v), as an event or condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to: (C) control the release of radioactive material and (D) mitigate the consequences of an accident.

There are no regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact Mr. Jonathan Johnson, Site Licensing Manager, at (423) 843-8129.

Respectfully,



Anthony L. Williams
Site Vice President
Sequoyah Nuclear Plant

Enclosure: Licensee Event Report 50-327 and 50-328/2018-001-00
cc: NRC Regional Administrator – Region II
NRC Senior Resident Inspector – Sequoyah Nuclear Plant

NRC FORM 366 (04-2018)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104		EXPIRES: 03/31/2020												
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)										Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov , and to the Desk Officer, Office of Information and Regulatory Affairs, NE0B-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.									
1. Facility Name Sequoyah Nuclear Plant Unit 1					2. Docket Number 05000327			3. Page 1 OF 7											
4. Title Inadequate Post Maintenance Testing Results in Condition Prohibited by Technical Specifications																			
5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved										
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name Sequoyah Nuclear Plant Unit 2		Docket Number 05000328								
06	26	2018	2018	- 001	- 00	09	18	2018	Facility Name N/A		Docket Number 05000								
9. Operating Mode 1		11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)																	
		☐ 20.2201(b)		☐ 20.2203(a)(3)(i)		☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)											
		☐ 20.2201(d)		☐ 20.2203(a)(3)(ii)		☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)											
		☐ 20.2203(a)(1)		☐ 20.2203(a)(4)		☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)											
		☐ 20.2203(a)(2)(i)		☐ 50.36(c)(1)(i)(A)		☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)											
		10. Power Level 100		☐ 20.2203(a)(2)(ii)		☐ 50.36(c)(1)(ii)(A)		☐ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)									
				☐ 20.2203(a)(2)(iii)		☐ 50.36(c)(2)		☐ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)									
				☐ 20.2203(a)(2)(iv)		☐ 50.46(a)(3)(ii)		☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)									
☐ 20.2203(a)(2)(v)				☐ 50.73(a)(2)(i)(A)		☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(ii)											
☐ 20.2203(a)(2)(vi)				☒ 50.73(a)(2)(i)(B)		☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(iii)											
		☐ 50.73(a)(2)(i)(C)		☐ Other (Specify in Abstract below or in NRC Form 366A)															
12. Licensee Contact for this LER																			
Licensee Contact Scott Bowman								Telephone Number (Include Area Code) 423-843-6910											
13. Complete One Line for each Component Failure Described in this Report																			
Cause	System	Component	Manufacturer	Reportable To ICES	Cause	System	Component	Manufacturer	Reportable To ICES										
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A										
14. Supplemental Report Expected					15. Expected Submission Date			Month	Day	Year									
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No								N/A	N/A	N/A									
Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)																			
On June 26, 2018, at 0902 eastern daylight time, a flow switch electronic circuit board associated with the B-train Emergency Gas Treatment System (EGTS) Air Cleanup Subsystem relative humidity heaters was replaced due to the circuit board approaching the end of its environmentally qualified life. Following board replacement, the flow switch was calibrated and the B-train of EGTS was returned to service in the standby condition. The post maintenance testing (PMT) did not require or recommend operability testing. On July 22, 2018, Operations personnel performed a surveillance instruction for Surveillance Requirement (SR) 3.6.10.1 and identified that the B-train relative humidity heaters did not energize as required by the SR. Operators declared Limiting Condition for Operation 3.6.10, EGTS Air Cleanup Subsystem, not met and entered Condition A for one EGTS Air Cleanup Subsystem train being inoperable. On July 23, 2018, the flow switch circuit board for the B-train EGTS relative humidity heaters, that had previously been removed on June 26, was reinstalled. Following reinstallation of the board, the B-train of EGTS was able to meet SR 3.6.10.1 and was declared operable July 24, 2018, at 0115. The cause of the event was the lack of formality in determining PMTs and return to operability (RTO) tests even though the procedure provides adequate guidance for this determination. The corrective action is to conduct information sharing with each Operations Crew and the Work Planners to determine PMT and RTO requirements in accordance with the maintenance procedure.																			

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER		
Sequoyah Nuclear Plant Unit 1	05000-327	YEAR 2018	SEQUENTIAL NUMBER - 001	REV NO. - 00

NARRATIVE**I. Plant Operating Conditions Before the Event**

At the time of the event, Sequoyah Nuclear Plant (SQN) Unit 1 and Unit 2 were in Mode 1 at 100 percent rated thermal power.

II. Description of Event**A. Event Summary:**

On June 26, 2018, a flow switch [EIS: FS] electronic circuit board [EIS: ECBD] associated with the B-train Emergency Gas Treatment System (EGTS) [EIS: BH] Air Cleanup Subsystem relative humidity heaters was replaced due to the circuit board approaching the end of its environmentally qualified life. Following board replacement, the flow switch was calibrated and the B-train of EGTS was returned to service in the standby condition. The post maintenance testing (PMT) did not require or recommend operability testing.

On July 22, 2018, Operations personnel performed a Technical Specification (TS) 3.6.10, EGTS Air Cleanup Subsystem, surveillance instruction (SI) for Surveillance Requirement (SR) 3.6.10.1 (operate each EGTS Air Cleanup Subsystem train for greater than or equal to 15 continuous minutes with heaters operating) and identified that the B-train relative humidity heaters did not energize as required by the SR. Operators declared Limiting Condition for Operation (LCO) 3.6.10 not met and entered TS 3.6.10 Condition A for one EGTS Air Cleanup Subsystem train being inoperable.

On July 23, 2018, the flow switch circuit board for the B-train EGTS relative humidity heaters, that had previously been removed on June 26, was reinstalled. Following reinstallation of the board, the B-train of EGTS was able to meet SR 3.6.10.1 and was declared operable July 24, 2018, at 0115 eastern daylight time (EDT).

A past operability evaluation (POE) determined that the B-train of the EGTS Air Cleanup Subsystem had been inoperable from June 26, 2018, at 0902 until July 24, 2018 at 0115. TS 3.6.10, Condition A allows 7 days to restore one train of the EGTS Air Cleanup Subsystem to operable status. TS 3.6.10 Condition B requires the unit to be in Mode 3 within 6 hours if the Required Action and associated Completion Time for Condition A are not met. Because one train of the EGTS Air Cleanup Subsystem was inoperable for greater than 174 hours (7 days plus 6 hours) without entering Mode 3, this event is reportable in accordance with 10 CFR 50.73(a)(2)(i)(B) as any operation or condition which was prohibited by the plant's Technical Specifications (TSs).

Additionally, the POE identified that the A-train of the EGTS Air Cleanup Subsystem had been tagged out of service for planned maintenance activities and declared inoperable concurrent with the B-train inoperability. Both trains of EGTS were inoperable for approximately 20 hours. With two trains of EGTS inoperable, LCO 3.0.3 should have been entered and the units should have



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transitioned to Mode 3 within 7 hours. This constitutes another event of an operation or condition which was prohibited by the plant's TSs. This event is also reportable in accordance with 10 CFR 50.73(a)(2)(v) as an event or condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to: (C) control the release of radioactive material and (D) mitigate the consequences of an accident.

- B. Status of structures, components, or systems that were inoperable at the start of the event and contributed to the event:

An operability test of the EGTS Air Cleanup Subsystem was not performed after maintenance. This allowed the B-train of EGTS to be inoperable longer than allowed by TSs without being detected.

- C. Dates and approximate times of occurrences:

Date/Time (EDT)	Description
February 2018	The work order for replacing the flow switch electronic circuit board was prepared and reviewed. The only post maintenance test (PMT) specified in the work order was to complete a calibration on the replaced board. The work planner and senior reactor operator who reviewed the work order failed to identify the need to functionally test the B-train EGTS Air Cleanup Subsystem after board replacement.
06/26/18, 0902	Both units enter TS 3.6.10 Condition A due to removing B-train EGTS from service for relative humidity heaters flow switch circuit board replacement and calibration.
06/26/18, 1443	Both units exit TS 3.6.10 Condition A after circuit board replacement and calibration. B-train EGTS was returned to service and placed in a standby condition. The B-train relative humidity heaters were not verified to be able to energize following circuit board replacement. The B-train of EGTS was still inoperable.
07/10/18, 0211	Both units enter TS 3.6.10 Condition A due to tagging the A-train of EGTS out of service for a maintenance outage.
07/10/18, 2200	Both units exit TS 3.6.10 Condition A due to A-train of EGTS being restored to operable status.



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Date/Time (EDT)	Description
07/22/18, 1008	Both units enter TS 3.6.10 Condition A due to removing B-train of EGTS from service to perform a B-train EGTS flow test associated with SR 3.6.10.1. Main Control Room operators discovered that the B-train relative humidity heaters were not energized.
07/24/18, 0115	Following replacement of the B-train EGTS relative humidity heaters flow switch circuit board, with the board removed on June 26, the B-train of EGTS was able to pass the flow test associated with SR 3.6.10.1. Both units exited TS 3.6.10 Condition A due to the EGTS Air Cleanup Subsystem being restored to operable status.

D. Manufacturer and model number of each component that failed during the event:

There was no component that failed during the event.

E. Other systems or secondary functions affected:

There were no other systems or secondary functions affected by this event.

F. Method of discovery of each component or system failure or procedural error:

Operations personnel performed an SI for SR 3.6.10.1 (operate each EGTS Air Cleanup Subsystem train for greater than or equal to 15 continuous minutes with heaters operating) and identified that the B-train relative humidity heaters did not energize.

G. Failure mode, mechanism, and effect of each failed component:

There was no component that failed during the event.

H. Operator actions:

Upon failure to meet SR 3.6.10.1, Operators declared LCO 3.6.10, EGTS Air Cleanup Subsystem, not met and entered TS 3.6.10 Condition A for one EGTS Air Cleanup Subsystem train being inoperable.

I. Automatically and manually initiated safety system responses:

There were no automatic or manually initiated safety system responses associated with this event.

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III. Cause of the Event**A. Cause of each component or system failure or personnel error:**

The cause of the event was the lack of formality in determining PMTs and return to operability (RTO) tests even though the procedure provides adequate guidance for this determination. PMTs may include RTOs and other maintenance tests. Work planners assumed that if no SR for a maintenance task could be found then no RTO test is required. Operations personnel did not provide enough vigor and challenge in their reviews of PMTs for work orders associated with safety related equipment. The extent of required review of RTO testing by Operation personnel was not fully understood leading to assumptions, in which for PMTs, may not be adequate to verify operability of equipment being tested.

B. Cause(s) and circumstances for each human performance related root cause:

There was a lack of formality in determining required testing following maintenance by the Work Management and Operations organizations.

The personnel involved were SQN employees. They were performing work associated with the replacement of the B-train EGTS Air Cleanup Subsystem flow switch electronic circuit card. There were no time or situational pressures present.

IV. Analysis of the Event:

The design bases for the EGTS are (1) to keep the air pressure within each Shield Building annulus below atmospheric at all times in which the integrity of that particular containment is required, and (2) to reduce the concentration of radioactive nuclides in annulus air that is released to the environs during a loss of coolant accident (LOCA) in either reactor unit to levels sufficiently low to keep the exclusion area boundary dose rate below the 10 CFR 100 guideline values.

The specified function of the relative humidity heaters is to reduce the relative humidity of the air stream to 70 percent relative humidity to prevent degradation of charcoal adsorber media efficiency. An engineering calculation determined the maximum relative humidity in the annulus resulting from a loss of coolant accident or a high energy line break would conservatively not exceed 61 percent relative humidity. Therefore, the Updated Final Safety Analysis Report specified function of the relative humidity heaters would be maintained with the heaters de-energized.

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V. Assessment of Safety Consequences

There were no actual safety consequences as a result of this event. The SQN probabilistic risk assessment model does not credit the use of the EGTS when calculating the probability of core damage or large early release. Accordingly, the risk associated with this event is considered to be very small.

- A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event:

Although B-train relative humidity heaters failed to energize, an engineering evaluation determined the B-train of EGTS was fully capable of meeting its specified function with the heaters not energized. Additionally, the A-train of EGTS was operable for the majority of the time the B-train was inoperable.

- B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident:

The event did not occur when the reactor was shut down.

- C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from discovery of the failure until the train was returned to service:

Operators identified a failure of B-train EGTS to meet SR 3.6.10.1 on July 22 at 1008. B-train EGTS was declared operable approximately 40 hours later on July 24 at 0115.

VI. Corrective Actions

This event was entered into the Tennessee Valley Authority Corrective Action Program under Condition Reports 1433166 and 1436225.

- A. Immediate Corrective Actions:

None.

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B. Corrective Actions to Prevent Recurrence or to reduce probability of similar events occurring in the future:

The corrective action is to conduct information sharing with each Operations Crew and the Work Planners to determine PMT and RTO requirements in accordance with the maintenance procedure.

Additionally, the preventive maintenance instruction associated with the work order was revised to ensure the requirements of the SI are met when performing future maintenance.

VII. Previous Similar Events at the Same Site:

A review of SQN LERs identified two previous events associated with human errors. LER 327/2016-005-00 reports a condition prohibited by TSs associated with the Hydrogen Mitigation System caused most likely by human error. LER 327/2016-006-00 reports a condition prohibited by TSs associated with the improper calibration of reactor trip instrumentation caused by Instrumentation & Controls Maintenance Personnel not being trained for operation of a digital multi-meter which resulted in an incorrect impedance setting being used during calibration.

VIII. Additional Information

There is no additional information.

IX. Commitments:

There are no commitments.